DocuFlowDB Database Schema Reference

> **Target Database**: `DocuFlowDB` (Microsoft SQL Server on `192.168.179.22:1443`)

> **Total Tables**: 45

> **Schema Definition Extraction Date**: 2026-10-09

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1. Authentication, Users & Access Control

Table: `users`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK** (Identity)	-
`user_uid`	`varchar(50)`	NULL	-	-
`employee_id`	`varchar(50)`	NOT NULL	-	-
`employee_name`	`varchar(150)`	NOT NULL	-	-
`name`	`varchar(150)`	NOT NULL	-	-
`username`	`varchar(150)`	NOT NULL	Unique identifier	-
`email`	`varchar(150)`	NOT NULL	Unique email address	-
`phone_number`	`varchar(20)`	NULL	Mobile number for SMS OTP	-
`division`	`varchar(100)`	NULL	Organizational Division	-
`department`	`varchar(100)`	NULL	Organizational Department	-
`plant`	`varchar(100)`	NULL	Plant/Location handle	-
`role`	`varchar(100)`	NULL	Legacy role name	-
`password_hash`	`varchar(255)`	NOT NULL	Bcrypt hashed password	-
`is_active`	`bit`	NULL	Active status flag	-
`is_deleted`	`bit`	NOT NULL	Soft-deletion flag	-
`deleted_at`	`datetime`	NULL	Soft-deletion timestamp	-
`mfa_enabled`	`bit`	NULL	MFA 2FA requirement flag	-
`mfa_type`	`varchar(50)`	NULL	`EMAIL`, `SMS`, `AUTHENTICATOR`	-
`mfa_secret`	`varchar(100)`	NULL	TOTP Base32 secret key	-
`last_login`	`datetime`	NULL	Last successful authentication	-
`active_session_id`	`varchar(100)`	NULL	Displaceable session identifier	-
`active_device_info`	`varchar(255)`	NULL	Device/User-Agent identifier	-
`session_created_at`	`datetime`	NULL	Session start timestamp	-
`last_activity_at`	`datetime`	NULL	Session heartbeat timestamp	-
`created_by`	`varchar(150)`	NULL	Creator username	-
`created_at`	`datetime`	NULL	Creation timestamp	-
`updated_at`	`datetime`	NULL	Last update timestamp	-
`role_id`	`int`	NULL	**FK** -> `roles.id`	-

Table: `roles`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`code`	`varchar(50)`	NOT NULL	Unique role code (e.g. `ADMIN`, `APPROVER`)	-
`name`	`varchar(100)`	NOT NULL	Display name	-
`description`	`varchar(255)`	NULL	Description of role	-
`is_active`	`bit`	NOT NULL	Role active status	-
`created_at`	`datetime`	NULL	Created timestamp	-

Table: `permissions`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`code`	`varchar(100)`	NOT NULL	Permission code (e.g. `doc:approve`)	-
`name`	`varchar(150)`	NOT NULL	Display name	-
`module`	`varchar(50)`	NULL	Module name	-
`action`	`varchar(50)`	NULL	CRUD action	-
`created_at`	`datetime`	NULL	Created timestamp	-
`subpage`	`varchar(50)`	NULL	Subpage association	-

Table: `role_permissions`

Column	Type	Nullable	Constraints & References	Default
`role_id`	`int`	NOT NULL	**PK**, **FK** -> `roles.id`	-
`permission_id`	`int`	NOT NULL	**PK**, **FK** -> `permissions.id`	-

Table: `user_access_logs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`username`	`varchar(150)`	NOT NULL	Target user	-
`event_type`	`varchar(50)`	NOT NULL	`LOGIN`, `LOGIN_FAILED`, `LOGOUT`, `MFA`	-
`status`	`varchar(50)`	NOT NULL	`SUCCESS`, `FAILED`	-
`ip_address`	`varchar(50)`	NULL	Client IP	-
`user_agent`	`varchar(255)`	NULL	Client user agent	-
`failure_reason`	`varchar(255)`	NULL	Failure detail if unsuccessful	-
`timestamp`	`datetime`	NULL	Event timestamp	-

Table: `otp_verifications`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`user_id`	`int`	NOT NULL	**FK** -> `users.id`	-
`otp_hash`	`varchar(255)`	NOT NULL	Bcrypt hash of OTP	-
`attempts`	`int`	NULL	Failed guess counter	-
`is_used`	`bit`	NULL	Consumption flag	-
`expires_at`	`datetime`	NOT NULL	Expiry datetime	-
`created_at`	`datetime`	NULL	Dispatch datetime	-

2. Document Ingestion & Invoice Processing

Table: `documents`

Column	Type	Nullable	Constraints & References	Default
`id`	`varchar(100)`	NOT NULL	**PK** (Document UUID / Key)	-
`invoice_number`	`varchar(100)`	NOT NULL	Document / Invoice identifier	-
`vendor_name`	`varchar(255)`	NOT NULL	Vendor or Counterparty	-
`vendor_code`	`varchar(100)`	NULL	ERP Vendor Master Code	-
`invoice_date`	`varchar(50)`	NULL	Date on invoice	-
`due_date`	`varchar(50)`	NULL	Due payment date	-
`po_number`	`varchar(100)`	NULL	Purchase Order number	-
`total_amount`	`decimal(18,2)`	NULL	Total monetary amount	-
`tax_amount`	`decimal(18,2)`	NULL	Tax component	-
`currency`	`varchar(10)`	NULL	ISO Currency code (`INR` , `USD`)	-
`category`	`varchar(100)`	NULL	Category classification	-
`payment_terms`	`varchar(100)`	NULL	Terms of payment	-
`status`	`varchar(50)`	NULL	Lifecycle status (`Pending Approval` , `Approved` , `Unrouted (No Rule Matched)` , etc.)	-
`workflow_stage`	`varchar(100)`	NULL	Current workflow stage (e.g. `Stage 1`)	-
`assigned_approver`	`varchar(150)`	NULL	Username / Email of active approver	-
`assigned_role`	`varchar(100)`	NULL	Target role for current stage	-
`file_name`	`varchar(255)`	NULL	Uploaded file name	-
`file_path`	`varchar(500)`	NULL	Relative storage path	-
`file_type`	`varchar(50)`	NULL	MIME / Extension	-
`file_size`	`int`	NULL	File size in bytes	-
`file_hash`	`varchar(64)`	NULL	SHA-256 integrity hash	-
`original_invoice_id`	`varchar(100)`	NULL	Previous revision ID if re-submitted	-
`rejection_reason`	`varchar(500)`	NULL	Reason for return/rejection	-
`line_items`	`varchar(max)`	NULL	Legacy JSON line items	-
`notes`	`varchar(max)`	NULL	Comments / internal notes	-
`audit_history`	`varchar(max)`	NULL	Structured audit history log	-
`extracted_data`	`varchar(max)`	NULL	Raw OCR / LLM extraction output	-
`created_at`	`datetime`	NULL	Ingestion timestamp	-

Column	Type	Nullable	Constraints & References	Default
`updated_at`	`datetime`	NULL	Modification timestamp	-
`document_type`	`varchar(100)`	NULL	Document taxonomy (`INVOICE` , `PRJ` , `FEEDBACK`)	-
`source_system`	`varchar(100)`	NULL	Origin (`MANUAL_UPLOAD` , `SAP` , `ORACLE` , `N8N`)	-
`priority`	`varchar(20)`	NULL	`Low` , `Medium` , `High` , `Urgent`	-
`department`	`varchar(100)`	NULL	Department	-
`division`	`varchar(100)`	NULL	Division handle	-
`plant`	`varchar(100)`	NULL	Plant handle	-
`location`	`varchar(100)`	NULL	Location code	-
`sub_department`	`varchar(100)`	NULL	Sub-department classification	-
`workflow_code`	`varchar(100)`	NULL	Associated workflow profile code	-
`version`	`int`	NULL	Document version number	-
`assigned_approver_id`	`int`	NULL	**FK** -> `users.id`	-
`source_id`	`int`	NULL	**FK** -> `source_records.id`	-
`document_key`	`varchar(200)`	NULL	Unique idempotency/business key	-
`is_latest_version`	`bit`	NULL	Active version flag	-
`is_locked`	`bit`	NULL	Concurrency lock flag	-
`locked_by`	`varchar(150)`	NULL	User holding edit lock	-
`locked_at`	`datetime`	NULL	Timestamp lock acquired	-

Table: `document_line_items`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`invoice_id`	`varchar(100)`	NOT NULL	**FK** -> `documents.id`	-
`line_number`	`int`	NOT NULL	Item sequence number	-
`item_code`	`varchar(100)`	NULL	SKU or ERP material code	-
`description`	`varchar(500)`	NOT NULL	Line description	-
`quantity`	`decimal(18,4)`	NULL	Line quantity	-
`unit_price`	`decimal(18,4)`	NULL	Unit price	-
`total_amount`	`decimal(18,2)`	NOT NULL	Line total amount	-
`tax_amount`	`decimal(18,2)`	NULL	Line tax amount	-
`cost_center`	`varchar(100)`	NULL	Cost center code	-
`gl_account`	`varchar(100)`	NULL	General Ledger account	-
`created_at`	`datetime`	NULL	Ingestion timestamp	-

Table: `document_locks`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`document_id`	`varchar(100)`	NOT NULL	**FK** -> `documents.id` (Unique)	-
`locked_by`	`varchar(150)`	NOT NULL	Username holding lock	-
`locked_at`	`datetime`	NOT NULL	Lock acquisition time	-
`expires_at`	`datetime`	NOT NULL	Lock expiration time	-

Table: `document_approval_logs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`document_id`	`varchar(100)`	NOT NULL	**FK** -> `documents.id`	-
`stage`	`varchar(100)`	NOT NULL	Stage approved/acted upon	-
`action`	`varchar(50)`	NOT NULL	`Approve`, `Reject`, `Return`, `Escalate`, `Hold`	-
`acted_by`	`varchar(150)`	NOT NULL	Acting username	-
`acted_at`	`datetime`	NOT NULL	Action timestamp	-
`comments`	`varchar(max)`	NULL	Comments or reasoning	-
`previous_status`	`varchar(50)`	NULL	Status prior to transition	-
`new_status`	`varchar(50)`	NULL	Status post-transition	-

Table: `document_type_field_configurations`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`document_type`	`varchar(100)`	NOT NULL	Associated document taxonomy	-
`field_name`	`varchar(100)`	NOT NULL	Attribute key	-
`field_label`	`varchar(150)`	NOT NULL	UI display label	-
`field_type`	`varchar(50)`	NOT NULL	`TEXT`, `NUMBER`, `DATE`, `DROPDOWN`	-
`is_required`	`bit`	NOT NULL	Mandatory validation rule	-
`is_readonly`	`bit`	NOT NULL	Read-only enforcement	-
`sort_order`	`int`	NULL	UI rendering order	-
`created_at`	`datetime`	NULL	Timestamp	-

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3. Workflow Profiles & Routing Rules

Table: `workflow_profiles`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`workflow_code`	`varchar(100)`	NOT NULL	Unique routing identifier	-
`name`	`varchar(200)`	NOT NULL	Profile display title	-
`description`	`varchar(500)`	NULL	Profile narrative description	-
`document_type`	`varchar(100)`	NOT NULL	Target taxonomy (`INVOICE` , `PRJ` , etc.)	-
`division`	`varchar(100)`	NULL	Scope division	-
`department`	`varchar(100)`	NULL	Scope department	-
`is_active`	`bit`	NOT NULL	Active state	-
`version`	`int`	NOT NULL	Profile version counter	-
`created_by`	`varchar(150)`	NULL	Author username	-
`created_at`	`datetime`	NULL	Creation timestamp	-
`updated_at`	`datetime`	NULL	Modification timestamp	-

Table: `workflow_step_definitions`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`workflow_id`	`int`	NOT NULL	**FK** -> `workflow_profiles.id`	-
`step_order`	`int`	NOT NULL	Stage execution order (1, 2, 3...)	-
`step_name`	`varchar(100)`	NOT NULL	e.g. `HOD Approval` , `Finance Review`	-
`approver_type`	`varchar(50)`	NOT NULL	`USER` , `ROLE` , `DYNAMIC_MANAGER`	-
`assigned_user_id`	`int`	NULL	**FK** -> `users.id`	-
`assigned_role`	`varchar(100)`	NULL	Assigned role title	-
`escalation_hours`	`int`	NULL	SLA escalation threshold in hours	-
`escalation_user_id`	`int`	NULL	**FK** -> `users.id`	-
`can_delegate`	`bit`	NOT NULL	Delegation authorization flag	-
`can_reject`	`bit`	NOT NULL	Rejection authorization flag	-
`can_return`	`bit`	NOT NULL	Return authorization flag	-
`created_at`	`datetime`	NULL	Creation timestamp	-

Table: `business_rules`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`name`	`varchar(200)`	NOT NULL	Rule display name	-
`description`	`varchar(500)`	NULL	Rule rationale	-
`document_type`	`varchar(100)`	NOT NULL	Target document taxonomy	-
`priority`	`int`	NOT NULL	Priority execution weight	-
`condition_json`	`varchar(max)`	NOT NULL	Dynamic JSON condition expressions	-
`workflow_id`	`int`	NOT NULL	**FK** -> `workflow_profiles.id`	-
`is_active`	`bit`	NOT NULL	Active state	-
`created_by`	`varchar(150)`	NULL	Author username	-
`created_at`	`datetime`	NULL	Created timestamp	-
`updated_at`	`datetime`	NULL	Updated timestamp	-

Table: `approval_assignments`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`document_id`	`varchar(100)`	NOT NULL	**FK** -> `documents.id`	-
`step_id`	`int`	NOT NULL	**FK** -> `workflow_step_definitions.id`	-
`user_id`	`int`	NOT NULL	**FK** -> `users.id`	-
`status`	`varchar(50)`	NOT NULL	`PENDING`, `APPROVED`, `REJECTED`, `SKIPPED`	-
`delegated_to_id`	`int`	NULL	**FK** -> `users.id`	-
`assigned_at`	`datetime`	NOT NULL	Timestamp assigned	-
`completed_at`	`datetime`	NULL	Timestamp acted upon	-

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4. Checklist & Validation Engine

Table: `checklist_templates`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`template_code`	`varchar(100)`	NOT NULL	Unique template identifier	-
`name`	`varchar(200)`	NOT NULL	Checklist title	-
`description`	`varchar(500)`	NULL	Description	-
`document_type`	`varchar(100)`	NOT NULL	Document taxonomy association	-
`items_json`	`varchar(max)`	NOT NULL	JSON checklist item questions & definitions	-
`is_active`	`bit`	NOT NULL	Active flag	-
`created_at`	`datetime`	NULL	Created timestamp	-
`updated_at`	`datetime`	NULL	Updated timestamp	-

Table: `checklist_rules`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`name`	`varchar(200)`	NOT NULL	Rule title	-
`template_id`	`int`	NOT NULL	**FK** -> `checklist_templates.id`	-
`document_type`	`varchar(100)`	NOT NULL	Target taxonomy	-
`condition_json`	`varchar(max)`	NOT NULL	JSON evaluation logic	-
`priority`	`int`	NOT NULL	Rule priority	-
`is_active`	`bit`	NOT NULL	Active flag	-
`created_at`	`datetime`	NULL	Created timestamp	-

Table: `document_checklist_states`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`document_id`	`varchar(100)`	NOT NULL	**FK** -> `documents.id`	-
`template_id`	`int`	NOT NULL	**FK** -> `checklist_templates.id`	-
`item_key`	`varchar(100)`	NOT NULL	Unique item key within checklist	-
`is_checked`	`bit`	NOT NULL	Checked status	-
`checked_by`	`varchar(150)`	NULL	User who checked the item	-
`checked_at`	`datetime`	NULL	Timestamp of check	-
`notes`	`varchar(500)`	NULL	Discrepancy or explanation notes	-

5. Generic & Project Requests

Table: `request_types`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`code`	`varchar(50)`	NOT NULL	Unique code	-
`name`	`varchar(100)`	NOT NULL	Display name	-
`description`	`varchar(500)`	NULL	Description	-
`is_active`	`bit`	NOT NULL	Status flag	-
`created_at`	`datetime`	NULL	Created timestamp	-

Table: `generic_requests`

Column	Type	Nullable	Constraints & References	Default
`id`	`varchar(100)`	NOT NULL	**PK**	-
`request_number`	`varchar(100)`	NOT NULL	Human-readable tracking number	-
`request_type_id`	`int`	NOT NULL	**FK** -> `request_types.id`	-
`title`	`varchar(255)`	NOT NULL	Summary title	-
`description`	`varchar(max)`	NULL	Detailed request content	-
`requester_id`	`int`	NOT NULL	**FK** -> `users.id`	-
`department`	`varchar(100)`	NULL	Department	-
`division`	`varchar(100)`	NULL	Division	-
`priority`	`varchar(20)`	NULL	Priority	-
`status`	`varchar(50)`	NOT NULL	Approval status	-
`current_stage`	`varchar(100)`	NULL	Active stage	-
`created_at`	`datetime`	NULL	Created timestamp	-
`updated_at`	`datetime`	NULL	Updated timestamp	-

Table: `generic_request_attachments`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`request_id`	`varchar(100)`	NOT NULL	**FK** -> `generic_requests.id`	-
`file_name`	`varchar(255)`	NOT NULL	File name	-
`file_path`	`varchar(500)`	NOT NULL	File path	-
`file_size`	`int`	NULL	Size in bytes	-
`uploaded_at`	`datetime`	NULL	Upload timestamp	-

Table: `generic_request_approval_logs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`request_id`	`varchar(100)`	NOT NULL	**FK** -> `generic_requests.id`	-
`stage`	`varchar(100)`	NOT NULL	Stage acted upon	-
`action`	`varchar(50)`	NOT NULL	Action taken	-
`acted_by`	`varchar(150)`	NOT NULL	Actor username	-
`comments`	`varchar(max)`	NULL	Comments	-
`timestamp`	`datetime`	NULL	Timestamp	-

Table: `project_requests`

Column	Type	Nullable	Constraints & References	Default
`id`	`varchar(100)`	NOT NULL	**PK**	-
`project_code`	`varchar(100)`	NOT NULL	Project tracking code	-
`title`	`varchar(255)`	NOT NULL	Project title	-
`description`	`varchar(max)`	NULL	Project scope	-
`budget_amount`	`decimal(18,2)`	NULL	Total approved budget	-
`currency`	`varchar(10)`	NULL	Currency	-
`project_manager_id`	`int`	NOT NULL	**FK** -> `users.id`	-
`division`	`varchar(100)`	NULL	Division handle	-
`department`	`varchar(100)`	NULL	Department handle	-
`start_date`	`datetime`	NULL	Estimated start	-
`end_date`	`datetime`	NULL	Estimated completion	-
`status`	`varchar(50)`	NOT NULL	Lifecycle status	-
`created_at`	`datetime`	NULL	Creation timestamp	-
`updated_at`	`datetime`	NULL	Update timestamp	-

Table: `project_request_attachments`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`project_id`	`varchar(100)`	NOT NULL	**FK** -> `project_requests.id`	-
`file_name`	`varchar(255)`	NOT NULL	File name	-
`file_path`	`varchar(500)`	NOT NULL	File path	-
`uploaded_at`	`datetime`	NULL	Upload timestamp	-

Table: `project_request_deliverables`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`project_id`	`varchar(100)`	NOT NULL	**FK** -> `project_requests.id`	-
`title`	`varchar(200)`	NOT NULL	Deliverable title	-
`description`	`varchar(max)`	NULL	Scope description	-
`due_date`	`datetime`	NULL	Due date	-
`status`	`varchar(50)`	NULL	Deliverable status	-

Table: `project_request_milestones`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`project_id`	`varchar(100)`	NOT NULL	**FK** -> `project_requests.id`	-
`milestone_name`	`varchar(200)`	NOT NULL	Milestone title	-
`target_date`	`datetime`	NULL	Target milestone date	-
`completed_date`	`datetime`	NULL	Actual completion date	-
`status`	`varchar(50)`	NULL	Milestone status	-

Table: `project_request_risks`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`project_id`	`varchar(100)`	NOT NULL	**FK** -> `project_requests.id`	-
`risk_description`	`varchar(max)`	NOT NULL	Risk explanation	-
`severity`	`varchar(20)`	NULL	`Low` , `Medium` , `High` , `Critical`	-
`mitigation_plan`	`varchar(max)`	NULL	Mitigation actions	-

Table: `project_request_approval_logs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`project_id`	`varchar(100)`	NOT NULL	**FK** -> `project_requests.id`	-
`stage`	`varchar(100)`	NOT NULL	Stage acted upon	-
`action`	`varchar(50)`	NOT NULL	Action taken	-
`acted_by`	`varchar(150)`	NOT NULL	Actor username	-
`comments`	`varchar(max)`	NULL	Notes / comments	-
`timestamp`	`datetime`	NULL	Timestamp	-

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6. Integration, Data Sync & Third-Party Webhooks

Table: `source_systems`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`code`	`varchar(50)`	NOT NULL	System code (`SAP_ERP` , `ORACLE` , `N8N`)	-
`name`	`varchar(100)`	NOT NULL	System display name	-
`description`	`varchar(255)`	NULL	Description	-
`is_active`	`bit`	NOT NULL	Active state	-
`created_at`	`datetime`	NULL	Created timestamp	-

Table: `source_records`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`source_system_id`	`int`	NOT NULL	**FK** -> `source_systems.id`	-
`source_primary_key`	`varchar(100)`	NOT NULL	Unique primary key in source ERP	-
`record_type`	`varchar(50)`	NOT NULL	e.g. `INVOICE` , `PURCHASE_ORDER`	-
`current_version_id`	`int`	NULL	Current revision ID	-
`created_at`	`datetime`	NULL	First ingested timestamp	-
`updated_at`	`datetime`	NULL	Last sync timestamp	-

Table: `source_record_versions`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`source_record_id`	`int`	NOT NULL	**FK** -> `source_records.id`	-
`version_number`	`int`	NOT NULL	Incrementing revision number	-
`raw_payload_json`	`varchar(max)`	NOT NULL	Exact JSON received during sync	-
`sync_run_id`	`int`	NULL	Associated sync run	-
`created_at`	`datetime`	NULL	Ingestion timestamp	-

Table: `sync_runs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`source_system_id`	`int`	NOT NULL	**FK** -> `source_systems.id`	-
`started_at`	`datetime`	NOT NULL	Batch start timestamp	-
`finished_at`	`datetime`	NULL	Batch end timestamp	-
`status`	`varchar(50)`	NOT NULL	`RUNNING`, `SUCCESS`, `FAILED`	-
`records_processed`	`int`	NULL	Record count	-
`records_failed`	`int`	NULL	Failure count	-
`error_log`	`varchar(max)`	NULL	Error diagnostics	-

Table: `integration_sync_logs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`endpoint`	`varchar(200)`	NOT NULL	API endpoint invoked	-
`source_ip`	`varchar(50)`	NULL	Client IP address	-
`status_code`	`int`	NOT NULL	HTTP response code returned	-
`request_body`	`varchar(max)`	NULL	Received payload	-
`response_body`	`varchar(max)`	NULL	Returned response	-
`timestamp`	`datetime`	NULL	Request timestamp	-

Table: `integration_audit_history`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`entity_type`	`varchar(50)`	NOT NULL	Target entity type	-
`entity_id`	`int`	NOT NULL	Entity ID	-
`action`	`varchar(50)`	NOT NULL	Modification action	-
`previous_value_json`	`varchar(max)`	NULL	Previous JSON snapshot	-
`new_value_json`	`varchar(max)`	NULL	New JSON snapshot	-
`changed_by`	`varchar(150)`	NULL	Actor	-
`timestamp`	`datetime`	NULL	Modification timestamp	-

Table: `third_party_applications`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`name`	`varchar(150)`	NOT NULL	Application display name	-
`code`	`varchar(50)`	NOT NULL	Application code	-
`description`	`varchar(max)`	NULL	Description	-
`base_url`	`varchar(500)`	NOT NULL	Application base URL	-
`environment`	`varchar(30)`	NULL	`PRODUCTION`, `STAGING`, etc.	-
`status`	`varchar(20)`	NULL	Application status	-
`auth_type`	`varchar(50)`	NULL	`BEARER_TOKEN`, `API_KEY`, `HMAC`	-
`auth_config_json`	`varchar(max)`	NULL	Auth parameter details	-
`created_at`	`datetime`	NULL	Created timestamp	-
`updated_at`	`datetime`	NULL	Updated timestamp	-

Table: `third_party_webhook_configs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`name`	`varchar(150)`	NOT NULL	Config name	-
`target_url`	`varchar(500)`	NOT NULL	Destination URL for webhooks	-
`auth_header_name`	`varchar(100)`	NULL	Custom auth header	-
`auth_token`	`varchar(500)`	NULL	Auth token	-
`hmac_secret`	`varchar(200)`	NULL	HMAC signature secret	-
`is_active`	`bit`	NULL	Active state	-
`events_json`	`varchar(max)`	NULL	Registered event subscriptions	-
`retry_count`	`int`	NULL	Max delivery retry count	-
`created_at`	`datetime`	NULL	Created timestamp	-
`updated_at`	`datetime`	NULL	Updated timestamp	-

Table: `callback_rules`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`rule_name`	`varchar(200)`	NOT NULL	Rule title	-
`description`	`varchar(max)`	NULL	Rule rationale	-
`application_id`	`int`	NOT NULL	**FK** -> `third_party_applications.id`	-
`status`	`varchar(20)`	NULL	Rule status	-
`priority`	`int`	NULL	Execution order	-
`trigger_event`	`varchar(100)`	NULL	`DOC_APPROVED`, `DOC_REJECTED`, etc.	-
`run_when`	`varchar(30)`	NULL	Trigger condition timing	-
`conditions_json`	`varchar(max)`	NULL	JSON rule conditions	-
`http_method`	`varchar(15)`	NULL	`POST`, `PUT`, `PATCH`	-
`url_mode`	`varchar(20)`	NULL	`BASE_URL_RELATIVE`, `CUSTOM`	-
`endpoint_path`	`varchar(500)`	NULL	Endpoint path	-
`custom_url`	`varchar(500)`	NULL	Full custom target URL	-
`body_type`	`varchar(30)`	NULL	Payload format	-
`payload_source`	`varchar(50)`	NULL	`DOC_JSON`, `TEMPLATE`, `SPROC`	-
`stored_procedure_name`	`varchar(200)`	NULL	Optional SQL stored procedure	-
`content_type`	`varchar(100)`	NULL	MIME Content-Type	-
`payload_mapping_json`	`varchar(max)`	NULL	Field mapping definitions	-
`raw_payload_template`	`varchar(max)`	NULL	Jinja/Mustache template	-
`query_params_json`	`varchar(max)`	NULL	URL query parameters	-
`headers_json`	`varchar(max)`	NULL	Custom HTTP request headers	-
`auth_override_type`	`varchar(50)`	NULL	Specific auth override	-
`auth_override_config_json`	`varchar(max)`	NULL	Override auth payload	-
`timeout_seconds`	`int`	NULL	HTTP request timeout	-
`success_criteria_json`	`varchar(max)`	NULL	HTTP response validation criteria	-
`follow_redirects`	`bit`	NULL	Follow 3xx redirects	-
`retry_config_json`	`varchar(max)`	NULL	Exponential backoff retry policies	-
`created_at`	`datetime`	NULL	Created timestamp	-
`updated_at`	`datetime`	NULL	Updated timestamp	-

Table: `callback_events`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`event_id`	`varchar(100)`	NOT NULL	Unique event UUID	-
`document_id`	`varchar(100)`	NOT NULL	**FK** -> `documents.id`	-
`rule_id`	`int`	NOT NULL	**FK** -> `callback_rules.id`	-
`application_id`	`int`	NOT NULL	**FK** -> `third_party_applications.id`	-
`source_primary_key`	`varchar(100)`	NULL	Source system primary key	-
`document_number`	`varchar(150)`	NULL	Document reference number	-
`decision`	`varchar(50)`	NOT NULL	Final approval decision	-
`status`	`varchar(50)`	NULL	`PENDING`, `DELIVERED`, `FAILED`	-
`attempt_count`	`int`	NULL	Dispatched attempts count	-
`max_attempts`	`int`	NULL	Retry limit	-
`next_retry_at`	`datetime`	NULL	Next scheduled delivery	-
`created_at`	`datetime`	NULL	Event creation timestamp	-
`updated_at`	`datetime`	NULL	Last event update	-

Table: `callback_attempts`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`callback_event_id`	`int`	NOT NULL	**FK** -> `callback_events.id`	-
`attempt_number`	`int`	NOT NULL	Sequence index of attempt	-
`http_method`	`varchar(15)`	NOT NULL	HTTP Method dispatched	-
`request_url`	`varchar(1000)`	NOT NULL	Target URL invoked	-
`request_headers_json`	`varchar(max)`	NULL	Dispatched request headers	-
`request_body`	`varchar(max)`	NULL	Dispatched request body	-
`response_status_code`	`int`	NULL	HTTP response code returned	-
`response_headers_json`	`varchar(max)`	NULL	Response headers received	-
`response_body`	`varchar(max)`	NULL	Response body received	-
`response_time_ms`	`int`	NULL	Round-trip latency in ms	-
`status`	`varchar(50)`	NULL	`SUCCESS`, `FAILURE`, `TIMEOUT`	-
`error_message`	`varchar(max)`	NULL	Detailed error traceback	-
`timestamp`	`datetime`	NULL	Execution timestamp	-

7. Notifications, RACI & System Administration

Table: `divisions`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`code`	`varchar(50)`	NOT NULL	Division code (e.g. `SPINNING`, `GARMENTS`)	-
`name`	`varchar(150)`	NOT NULL	Division display name	-
`is_active`	`bit`	NOT NULL	Active status flag	-
`created_at`	`datetime`	NULL	Creation timestamp	-

Table: `departments`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`code`	`varchar(50)`	NOT NULL	Department code (e.g. `FINANCE`, `HR`)	-
`name`	`varchar(150)`	NOT NULL	Department display name	-
`division_id`	`int`	NULL	**FK** -> `divisions.id`	-
`is_active`	`bit`	NOT NULL	Active status flag	-
`created_at`	`datetime`	NULL	Creation timestamp	-

Table: `notification_provider_configs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`smtp_server`	`varchar(200)`	NOT NULL	Hostname (e.g. `smtp.office365.com`)	-
`port`	`int`	NULL	SMTP port (e.g. `587`)	-
`username`	`varchar(200)`	NULL	SMTP username / sender account	-
`encrypted_password`	`varchar(200)`	NULL	Encrypted / plaintext SMTP password	-
`sender_email`	`varchar(200)`	NULL	Sender mailbox address	-
`sender_name`	`varchar(200)`	NULL	Sender display name	-

Table: `notification_raci_matrices`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`workflow_profile`	`varchar(200)`	NOT NULL	Target profile handle	-
`event_name`	`varchar(100)`	NOT NULL	`ON_SUBMIT`, `ON_APPROVE`, `ON_REJECT`	-
`responsible_emails`	`varchar(max)`	NULL	Responsible party emails	-
`accountable_emails`	`varchar(max)`	NULL	Accountable party emails	-
`consulted_emails`	`varchar(max)`	NULL	Consulted party emails	-
`informed_emails`	`varchar(max)`	NULL	Informed party emails	-
`title_template`	`varchar(500)`	NULL	Notification title template	-
`message_template`	`varchar(max)`	NULL	Notification body template	-

Table: `in_app_notifications`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`notification_id`	`varchar(100)`	NULL	UUID notification tracking handle	-
`document_id`	`varchar(100)`	NOT NULL	Associated document ID	-
`recipient_handle`	`varchar(200)`	NOT NULL	Recipient username or email	-
`notification_type`	`varchar(50)`	NULL	Type (`ASSIGNMENT`, `STATUS_CHANGE`)	-
`title`	`varchar(300)`	NOT NULL	Notification title	-
`message`	`varchar(max)`	NOT NULL	Notification message body	-
`is_read`	`bit`	NULL	Read flag	-
`created_at`	`datetime`	NULL	Dispatched timestamp	-

Table: `system_engine_logs`

Column	Type	Nullable	Constraints & References	Default
`id`	`int`	NOT NULL	**PK**	-
`module_name`	`varchar(100)`	NULL	Originating subsystem (`RULES`, `SYNC`, `OCR`)	-
`invoice_id`	`varchar(100)`	NULL	**FK** -> `documents.id`	-
`action`	`varchar(150)`	NOT NULL	Engine action executed	-
`user`	`varchar(150)`	NULL	Initiating user / system handle	-
`log_level`	`varchar(20)`	NULL	`INFO`, `WARNING`, `ERROR`	-
`details`	`varchar(max)`	NULL	Log text or stack trace	-
`timestamp`	`datetime`	NULL	Event timestamp	-